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# Decompressive Craniectomy for Traumatic Brain Injury: The Role of Cranioplasty and Hydrocephalus on Outcome

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■ OBJECTIVE: **After severe traumatic brain injury (sTBI) associated with uncontrollable high intracranial pressure (ICP), today the main challenge for neurosurgeons remains to identify who may obtain benefit from decompressive craniectomy (DC) and which factors after DC influence the prognosis of these patients. The aim of this paper was to identify the pre- and postoperative determinants of outcome after DC.**

■ METHODS: **This retrospective study included all patients undergoing DC for sTBI from 2003 to 2011. The 6-month outcome, assessed using the Glasgow Outcome Scale (GOS), was dichotomized into favorable (GOS scores 4 and 5) and unfavorable (GOS scores 1–3) outcome. Predictors of outcome were identified by uni- and multivariate analysis.**

■ RESULTS: **There were 190 patients who underwent DC for sTBI in this study. Sixty patients (31.6%) died within 30 days after DC. Independent prognostic factors for survival after 30 days were Glasgow Coma Scale score at admission greater than 5 ($P = 0.002$) and bilateral pupil reactivity ($P < 0.0001$). Thirty days after DC, 67 patients (51.5%) out of 130 had unfavorable outcome (GOS scores 1–3) and 63 patients (49.5%) presented favorable outcome (GOS scores 4 and 5). The independent preoperative prognostic factors for poor outcome were age over 65 years ($P < 0.0001$) and bilateral absence of pupil reactivity ($P = 0.0165$). After DC, onset of postoperative hydrocephalus and delayed cranioplasty (3 months after DC) was associated with unfavorable outcome at multivariate analysis ($P = 0.002$ and $P < 0.0001$, respectively).**

■ CONCLUSIONS: **In our study, the development of hydrocephalus after DC for sTBI and delayed cranial reconstruction were associated with unfavorable outcome.**

## INTRODUCTION

Uncontrollable high intracranial pressure (ICP) is the main cause of death after severe traumatic brain injury (sTBI).[1] Decompressive craniectomy (DC) has been proposed as a lifesaving treatment in patients with refractory high ICP.[2]

Two recent randomized controlled trials (Randomized Evaluation of Surgery with Craniectomy for Uncontrollable Elevation of Intracranial Pressure and DECompressive CRAniectomy trials) found that DC can decrease ICP value and mortality.[3,4] However, in these trials, DC was also associated with the increased rate of survival in patients in a vegetative state or with severe disability.[3-5] Still, today, there is a question about who may obtain benefit from the surgical decompression and which postoperative factors influence the prognosis of patients treated with DC after sTBI.[6] The identification of preoperative prognostic factors (e.g., age, bilateral pupil reactivity, Glasgow Come Scale [GCS] score at admission) improves the selection of patients who can benefit from DC after sTBI. For instance, the absence of bilateral pupillary response to light may be a sign of nefarious intracranial pathologies associated with bad outcome, including high rate of mortality.

**Key words**
- Cranioplasty
- Decompressive craniectomy
- Hydrocephalus
- Traumatic brain injury

**Abbreviations and Acronyms**
**DC**: Decompressive craniectomy
**GCS**: Glasgow Come Scale
**GOS**: Glasgow Outcome Scale
**GOS-E**: Extended Glasgow Outcome Scale
**ICP**: Intracranial pressure
**SDH**: Subdural hemorrhage
**sTBI**: Severe traumatic brain injury
**VPS**: Ventriculoperitoneal shunt

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Moreover, the role of postoperative conditions specifically associated with DC on outcome, such as hydrocephalus, subdural hygromas, increased incidence of contusions, evolution of preexisting intradural mass lesions, infections, and sinking brain syndrome, has not been fully investigated.[7-10] Finally, the real impact of cranioplasty on outcome in a previous series published about DC remains unclear.[11]

In the current study, we report the outcome after DC and analyze the impact on outcome of the aforementioned factors in 190 patients with sTBI referred to a large-level trauma center.

## MATERIALS AND METHODS

A retrospective study was conducted on patients treated with DC for sTBI at our department between January 2003 and December 2011.

Demographic characteristics (age and sex), mechanism of injury, timing between injury and DC, admission GCS score, admission bilateral pupil reactivity and size, brain computed tomography scan findings (subdural hematoma, brain contusion, etc.), Rotterdam score, and surgical information including type and reason for DC were recorded and analyzed. Data concerning postoperative complications, such as meningitis, hydrocephalus, subdural hygroma, incidence of new brain contusions, and expansion of contralateral posttraumatic mass lesions, were analyzed too. Finally, we evaluated the number of patients that underwent cranioplasty within 3 months.

The primary outcome was assessed by means of the 5-point Glasgow Outcome Scale (GOS) at 6-month follow-up: death, persistent vegetative state, and severe disability were assessed as unfavorable outcome (GOS scores 1–3), whereas GOS scores 4 and 5 (good recovery and moderate disability) were assessed as favorable outcome. The secondary outcome was mortality at 30 days.

The indications for DC were based on the current literature and the Brain Trauma Foundation's guidelines for management of ICP after traumatic brain injury.[2] Two cases were included: 1) evacuation of a mass lesion with evidence of brain swelling (primary DC) with contraindication to replace the bone flap, and 2) ICP refractory to medical treatment (secondary DC).

Unilateral or bifrontal DC was performed depending on the presence of a confined lesion or diffuse brain swelling.

Data were analyzed with Statistical Package for Social Sciences version 12 (SPSS Inc., Chicago, Illinois, USA). Continuous variables were compared using Student t test and $\chi^2$ test for discrete variables. The multiple logistic regression was used to identify independent pre- and postoperative risk factors (separately analyzed) associated with the outcome in the whole population. The outcome was analyzed as a dichotomous dependent variable: good outcome (good recovery and moderate disability according to the GOS) and poor outcome (severe disability, vegetative state, and death).

The Kaplan-Meyer survival analysis was used to evaluate the cumulative survival in the first 30 days. Determinants of 30-day mortality were analyzed with Cox regression model.

## RESULTS

Data of 190 patients (149 men and 41 women; mean age, 50 ± 15.5 years; range, 14–86 years) who underwent DC for sTBI in our center were included in this study. The main cause of sTBI was motor vehicle accident in 94 patients (49.5%) and accidental fall in 74 patients (39%). In the last 22 cases (11.5%), the trauma etiology

**Table 1.** Baseline Characteristics of 190 Patients with Severe Traumatic Brain Injury (sTBI) Undergoing Decompressive Craniectomy

| Variable | Value |
|---|---|
| Mean age (range) (years) | 50 (14–86) |
| <65 years | 120 (63.1) |
| ≥65 years | 70 (36.9) |
| Cause of sTBI | |
| Road traffic accidents | 111 (58.4) |
| Falls | 75 (39.6) |
| Other | 4 (2) |
| GCS score at admission | |
| 3–5 | 129 (67.9) |
| 6–8 | 59 (31.1) |
| >8 | 2 (1) |
| Bilateral pupil reactivity to light | |
| Yes | 156 (82.1) |
| No | 34 (17.9) |
| Extracranial injury | |
| No | 109 (57.4) |
| Yes | 81 (42.3) |
| Timing of DC | |
| Mean (hours) | 24.5 |
| <48 hours | 154 (81) |
| ≥48 hours | 36 (18.9) |
| Type of DC | |
| Primary | 109 (57.4) |
| Secondary | 81 (42.6) |
| Surgical approach | |
| Unilateral | 153 (80.5) |
| Bilateral | 37 (19.5) |
| Type of intracranial lesions | |
| Only acute subdural hematoma | 40 (21) |
| Only brain contusion | 46 (24.2) |
| Acute subdural he and brain contusion | 99 (52.2) |
| Diffuse brain swelling | 5 (2.6) |
| Rotterdam score | |
| 1–4 | 101 (53.2) |
| 5 and 6 | 89 (46.8) |

Values are number of patients (%) or as otherwise indicated.
GCS, Glasgow Coma Scale; DC, decompressive craniectomy.

**Table 2.** Univariate and Multivariate Analysis of Preoperative Factors Predicting 30-Day Mortality in Our Case Series (Relative Hazard) with 95% Confidence Interval

| | | | Univariate Analysis | Multivariate Analysis | | |
|---|---|---|---|---|---|---|
| **Variable** | **Number of Patients Dead** | **Number of Patients Alive** | ***P* Value** | ***P* Value** | **RH** | **95% CI** |
| Age | | | 0.0067 | | | |
| <65 years | 30 | 91 | | | | |
| >65 years | 30 | 29 | | | | |
| GCS score at admission | | | 0.0003 | 0.0022 | 0.2652 | 0.1138—0.6180 |
| 3—5 | 54 | 75 | | | | |
| >5 | 6 | 55 | | | | |
| Bilateral pupil reactivity | | | <0.0001 | 0.0001 | 0.2279 | 0.1348—0.3854 |
| Yes | 34 | 122 | | | | |
| No | 26 | 8 | | | | |
| Pupil abnormality | | | <0.0001 | | | |
| No | 10 | 72 | | | | |
| At least one pupil dilated | 50 | 58 | | | | |
| Rotterdam score | | | 0.006 | | | |
| 1—4 | 23 | 78 | | | | |
| 5 and 6 | 37 | 52 | | | | |

RH, relative hazard; CI, confidence interval; GCS, Glasgow Coma Scale.

was unknown. At admission, 57 patients (30%) had a GCS score of 3, 72 patients had a GCS score of 4 or 5 (37.6%), and 71 patients (37.4%) had a GCS score of greater or equal to 6. Pupils were not reactive bilateral to the light in 34 patients out of 190 (17.9%). Most of the patients had an intradural posttraumatic lesion: subdural hemorrhage (SDH) in 40 patients (21%), hemorrhagic contusion in 46 patients (24.2%), and SDH with contusions in 99 patients (52.2%). A diffuse brain swelling without focal lesion was documented in 5 cases.

Primary decompression with hematoma evacuation and without ICP monitoring was performed on 109 patients out of 190 (57.36%) and secondary decompression after untreatable ICP was treated on 81 out of 190 patients (42.64%).

The mean interval between admission and DC was 24.5 ± 2.6 hours (range, 1 hour to 10 days); 154 (81%) patients out of 190 underwent DC within 48 hours after admission. Unilateral DC was performed in 153 cases (80.5%) out of 190, whereas bifrontal DC was done in 37 (19.5%) patients.

Table 1 summarizes the baseline characteristics of the patients.

### Thirty-Day Mortality

The overall mortality rate was 46.8% (89/190 patients), and 60 patients (31.6%) died within 30 days after DC. The cumulative probability of survival within the 30 days after DC was 68%. The univariate analyses showed that age over 65 years old, lower GCS score at admission, absence of bilateral pupil reactivity, at least one dilated pupil, Rotterdam score, and type of lesion (SDH and brain contusion) were associated with higher risk of 30-day mortality (Table 2). Independent prognostic factors for survival after 30 days were GCS score at admission ($P = 0.002$) and bilateral pupil reactivity ($P < 0.0001$).

### GOS Score at 6 Months After DC

Among the 130 patients alive after 30 days after DC, outcome was poor (GOS severe disability, persistent vegetative state, and death) in 51.5% of patients (n = 67), whereas 48.5% of patients (n = 63) presented good outcome defined with GOS score of 4 (moderate disability; n = 41) or GOS score of 5 (good recovery; n = 22). In the group of 67 patients with poor outcome, 28 patients (41.8%) died (GOS score 1), 19 patients (28.3%) presented persistent vegetative status (GOS score 2), and 20 patients (29.9%) had severe disability (GOS score 3).

### Preoperative Predictors of Outcome

Table 3 reports preoperative factors predicting outcome. At the multivariate logistic regression, the independent prognostic factors for poor outcome were age over 65 years old ($P < 0.0001$) and bilateral absence of pupil reactivity ($P = 0.0165$).

### Postoperative Predictors of Outcome

Table 4 summarizes postoperative factors predicting outcome. At the multivariate logistic regression, the only independent factors associated with unfavorable outcome were onset of postoperative hydrocephalus ($P = 0.002$) and delayed cranial reconstruction (after 3 months) ($P < 0.0001$).

**Table 3.** Univariate and Multivariate Analysis of Preoperative Factors Predicting Favorable Outcome (Glasgow Outcome Scale Score of 4 or 5) in Our Case Series

| Variable | Number of Patients with GOS Score 1–3 | Number of Patients with GOS Score 4 and 5 | Univariate Analysis | Multivariate Analysis | | |
|---|---|---|---|---|---|---|
| | | | *P* Value | *P* Value | Odd Ratio | 95% CI |
| Age | | | 0.0001 | <0.0001 | 10.12 | 3.7639–27.2136 |
| <65 years | 36 | 55 | | | | |
| >65 years | 31 | 8 | | | | |
| Sex | | | 0.95 | | | |
| M | 55 | 51 | | | | |
| F | 12 | 12 | | | | |
| GCS score at admission | | | 0.16 | | | |
| 3–5 | 44 | 31 | | | | |
| >5 | 23 | 32 | | | | |
| Pupil reactivity | | | 0.08 | | | |
| Yes | 60 | 62 | | | | |
| No | 7 | 1 | | | | |
| Pupil abnormality | | | <0.39 | | | |
| No | 36 | 36 | | | | |
| At least one pupil dilated | 31 | 27 | | | | |
| Rotterdam score | | | 0.016 | | | |
| 1–4 | 44 | 22 | | | | |
| 5 and 6 | 23 | 41 | | | | |

GOS, Glasgow Outcome Scale; M, male; F, female; GCS, Glasgow Coma Scale.

### Complications of DC

The overall rate of complications was 46.8% (89/190 patients). Among these 89 patients, 38 (20%) required one or more reoperations for the treatment of complications. The most common complications of DC in our series are summarized in Table 5 and include symptomatic hemorrhagic progression of a contusion (n = 15; 7.8%), hydrocephalus (n = 37; 19.5%), subdural hygroma (n = 8; 4.2%), infections (n = 4; 2.1%), new ipsilateral hematoma (n = 14; 7.4%), new contralateral hematoma (n = 5; 2.6%), and another 6 complications.

### Hydrocephalus

Hydrocephalus developed in 37 out 130 patients (28.4%) after a median of 6.43 months after DC (range, 1–15 months); 29 out of 37 patients (16.2%) showed an interhemispheric hygroma in a previous computed tomography scan (P = 0.0067).

The diagnosis of hydrocephalus was derived from a combination of clinical signs and radiographic features. For some patients, no definite change in clinical examination was noted, but considerable enlargement of ventricles was noted on serial head computed tomography scans.

The development of hydrocephalus required a ventriculoperitoneal shunt (VPS) in 34 out of 37 patients (91.9%), whereas in the remaining 3 patients, cerebrospinal fluid hydrodynamics disturbances resolved after cranioplasty. Among the patients treated with a VPS, in 26 cases the shunt was implanted after cranioplasty, in 5 cases a concomitant VPS operation and cranioplasty were performed, and only 3 patients underwent VPS operation before cranial reconstruction.

### Cranioplasty

In this series, 59 out of 130 patients alive after 30 days after DC received cranioplasty with autologous bone (cryopreserved) or synthetic implant within 3 months after DC. In this subgroup of patients who received cranioplasty within 3 months, favorable outcome occurred in 81% of the cases (49/59 patients) compared with 19.7% of those (14/71 patients) in the group of patients treated with cranial reconstruction after 3 months (P < 0.0001).

## DISCUSSION

Uncontrollable high ICP is the main cause of death and disability after sTBI.[1-3] The placement of an ICP monitor in comatose sTBI patients with cerebral contusions is the standard care in developed countries.[1] Medical therapy may decrease ICP value, but when maximal medical treatment is not effective, DC represents the last tier treatment.[2]

**Table 4.** Univariate and Multivariate Analysis of Postoperative Factors Predicting Favorable Outcome (Glasgow Outcome Scale Score of 4 or 5) in Our Case Series

| Variable | Number of Patients with GOS Score 1–3 | Number of Patients with GOS Score 4 and 5 | Univariate Analysis | Multivariate Analysis | | |
|---|---|---|---|---|---|---|
| | | | *P* Value | *P* Value | Odd Ratio | 95% CI |
| Post-DC hydrocephalus | | | 0.0002 | <0.002 | 0.1299 | 0.0489–0.3447 |
| Yes | 29 | 8 | | | | |
| No | 38 | 55 | | | | |
| Cranioplasty within 3 months | | | | | | |
| Yes | 10 | 49 | <0.0001 | <0.0001 | 0.149 | 0.028–0.791 |
| No | 57 | 14 | | | | |

GOS, Glasgow Outcome Scale; CI, confidence interval; DC, decompressive craniectomy.

## Outcome After DC

In the last 30 years, a large number of retrospective series reported the use of DC as an effective measure to reduce ICP after sTBI.[6] More recently, 2 large multicentric prospective randomized controlled trials investigated the effect of DC on clinical outcomes in patients with refractory traumatic intracranial hypertension.[3,4]

The DECompressive CRAniectomy study analyzed the role of early bifrontal DC in patients with uncontrollable high ICP secondary to diffuse cerebral swelling.[4] ICP was considered uncontrollable after 15 minutes of values higher than 20 mm Hg.

The trial reported lower ICP values in decompressed patients rather than in patients treated with medical therapy. However, at 6-month follow-up, 70% of patients in the DC group had unfavorable outcome versus 51% of patients in the standard care group.

This trial has been criticized according to different biases (presence of diffuse injury, transient increase of ICP, surgical technique, etc.) that may have altered the results. However, the real value of this paper consisted in the fact that it highlighted an important issue: ICP decreasing does not always correlate with a better outcome.[9]

Also in our series, the relatively high proportions of overall mortality rate (46.8%) and unfavorable outcome (51.5%) reflected an inappropriate selection of patients in many cases. Furthermore, also in light of the high rate of complications of patients with DC in our study (35.2%), the indiscriminate use of surgical decompression appears not safe or effective.

## Outcome After DC and Cranioplasty

Recently, Hutchinson et al.[3] reported the results about the Randomized Evaluation of Surgery with Craniectomy for Uncontrollable Elevation of Intracranial Pressure trial. This trial compared DC with continued medical treatment for refractory intracranial hypertension after sTBI.

In this randomized controlled trial, 408 patients with refractory elevated ICP were enrolled for over 10 years (from 2004 to 2014). A ICP value above 25 mm Hg for a time between 1 and 12 hours was considered as uncontrollable high intracranial hypertension, and the patients were randomly assigned to undergo DC or be treated with medical therapy.

The primary outcome was assessed with the Extended Glasgow Outcome Scale (GOS-E) at 6 months. The trial showed better ICP control, lower mortality, higher rates of vegetative state, and higher severe disability in the decompressed group than in the medical one; the rates of moderate and good recovery were similar in the 2 groups.

The dichotomized GOS-E results (GOS-E score 4–8 was defined as favorable outcome, whereas GOS-E score 1–3 was defined as unfavorable outcome) did not show a higher percentage of patients with a favorable outcome in the surgical group than in the medical group at 6-month follow-up (42.8% and 34.6%, respectively; $P = 0.12$). On the contrary, there was a significant difference at 12-month follow-up (45.4% vs. 32.4%, respectively; $P = 0.01$).

This trial is very well designed; however, a possible limitation, highlighted by the authors themselves, is that the data on cranioplasty and the subsequent impact of cranial reconstruction on neurologic outcome were not evaluated. It is widely reported in the literature that cranial reconstruction improves cerebral perfusion and cerebrospinal fluid dynamics, usually altered by atmospheric

**Table 5.** Rates of Complications After Decompressive Craniectomy in Our Series

| Complications Associated with Decompressive Craniectomy | Number of Patients (%) |
|---|---|
| Hydrocephalus | 37 (19.5) |
| Subdural hygroma | 8 (4.2) |
| Infections | 4 (2.1) |
| Hemorrhagic progression of a contusion | 15 (7.8) |
| New ipsilateral hematoma | 14 (7.4) |
| New contralateral/remote hematoma | 5 (2.6) |
| Syndrome of the trephined | 6 (3.1) |

pressure in patients with open skull. Therefore, in clinical practice, it is often associated with improvement in neurologic status.[12,13] In this regard, a recent meta-analysis, including 8 retrospective observational studies, showed that the cranioplasty is associated with significant, quantifiable neurologic improvement. Further, early cranioplasty (within 90 days after DC) may lead to even greater improvements.[13]

In our series, the cranioplasty within 3 months had a major positive impact on neurologic outcome: in the subgroup of patients who received cranioplasty within 3 months, favorable outcomes occurred in 81% of the cases (49/59 patients) compared with 19.7% of the cases (14/71 patients) in the group of patients with cranial reconstruction after 3 months ($P < 0.0001$). In our opinion, the design of future trials about DC should take into account this issue and include a subgroup analysis of patients who underwent cranial reconstruction before the end point follow-up for a correct estimate of clinical outcome after DC.

### Pre- and Postoperative Prognostic Factors in Decompressed Patients

Nowadays, the question is whether there is a subset of patients who benefit from surgical decompression and which factors influence the prognosis of patients submitted to DC, not only before the procedure but also after DC. Many studies have documented that the principal effect of DC is the decrease of mortality and the increase of vegetative state or severely disabled patients.

In our study, the overall mortality rate was 46.8% (89/190 patients), and 60 patients (31.6%) died within 30 days after DC. Independent prognostic factors for survival after 30 days were only GCS score at admission greater than 5 ($P = 0.002$) and bilateral pupil reactivity ($P < 0.0001$).

In 67 out of 130 patients (51.5%) still alive after 30 days after surgery, the outcome was poor (GOS score 1–3), whereas 63 patients (48.5%) had a favorable outcome (GOS score 4 and 5).

At the multivariate logistic regression, the independent prognostic factors for poor outcome were age over 65 years ($P < 0.0001$), occurrence of postoperative hydrocephalus ($P < 0.0001$), and absence of bilateral pupil reactivity ($P = 0.0165$).

As reported in other studies, age over 65 years was an independent factor associated with unfavorable outcome.[6,13] The age limit in many retrospective and prospective studies is 60–65 years old, whereas in our series 70 out of 190 patients (36.9%) were aged older than 65 years.

These data contrast with the literature because at the beginning of our experience with DC in sTBI (years 2003–2007), the studies about the effect of age on outcome after DC were not so conclusive. However, after the publication of the retrospective series of 526 DCs by Tagliaferri et al.[6] in 2012, there were no longer doubts about the determining role of age regarding surgical indication: in this series, age was the most significant predictor factor of survival, together with pupil reactivity, time of decompression, and size of the bone flap. The effect of age in predicting survival was so important that in patients over 65 years of age, the authors did not find any other significant factor related to survival. Then, our study confirmed once again the importance of age as a negative prognostic factor.

Moreover, this issue is a critical point also because in high-income countries, epidemiologic patterns appear to be changing with an increasing incidence of traumatic brain injury in older adults.[14] Based on this finding, future prospective randomized studies about DCs in patients aged over 65 years must be designed to better analyze this issue. However, in current clinical practice, neurosurgeons must be aware that DC in elderly patients may be not effective in terms of final clinical outcome.

The literature reports that the absence of bilateral pupil reactivity at admission predicts a bad outcome in patients with sTBI, even in patients undergoing DC. In our study, pupil reactivity resulted in a preoperative predictor of outcome. Indeed, the absence of bilateral pupillary response to light is a sign of more nefarious intracranial pathologies associated with bad outcome, including high rate of mortality.

Postoperative hydrocephalus and subdural hygroma in sTBI patients may be caused by abnormal cerebrospinal fluid dynamics. De Bonis et al.[15] documented that craniotomies crossing the midline are associated with postoperative hydrocephalus and intracranial subdural hygroma. Recently, Khalili et al.[16] reported that postoperative hydrocephalus was associated with unfavorable outcome.

In our study, postoperative hydrocephalus was independently associated with unfavorable outcome; a possible explanation is that a delayed diagnosis of this condition impairs neurologic improvement during rehabilitation. Moreover, in these frail patients, surgical treatment of hydrocephalus by VPS is associated with the risk of surgical complications, including infections, shunt malfunctions, and hyperdrainage.

An exponential number of studies, commentaries, and reviews were published in the last years about DC.[1-12] Although we are aware that DC is able to reduce the mortality of patients with uncontrollable intracranial hypertension, on the other hand, it leads to an increase in patients with severe disability. In this scenario, the data provided by our series on outcome after DC are not new and confirmed this point.

Finally, the most important aspect of our study is that the development of hydrocephalus after DC and delayed cranial reconstruction were associated with unfavorable outcome. Therefore, the diagnosis and subsequent treatment of hydrocephalus after DC and early cranioplasty represented an important issue in the management of the decompressed patient.

For these reasons, decompressed patients should not be abandoned after surgery; they require constant monitoring by a team of neurosurgeons experienced in this field.

The main limitations of this study are as follows. First, we included both primary decompression (with hematoma evacuation without ICP monitoring) and secondary decompression (after untreatable ICP). These are 2 different types of patients with different prognoses and they could represent a possible confounding factor. Finally, the result of cranioplasty on outcome reported in this study could be partially affected by a bias related to the possible different severity of overall illness and health between the 2 analyzed groups.

## CONCLUSIONS

DC after sTBI was associated with an elevated number of survivors with severe neurologic disability. Independent prognostic factors

for survival after 30 days from DC were GCS score at admission greater than 5 and bilateral pupil reactivity.

On multivariate analysis, preoperative independent predictors of unfavorable outcome were age over 65 years and absence of bilateral pupil reactivity at admission. The identification of these factors can improve the selection of patients who may benefit from DC after sTBI.

After DC, the development of hydrocephalus and cranial reconstruction after 3 months were associated with unfavorable outcome.

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